YEAR 7 MATHS · CHAPTER 4 · MIND MAP

Algebraic techniques

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1 Introduction to algebra

A pronumeral is a letter for a number. $3x$ means $3 \times x$.

"7 more than n": $n + 7$

"Double y": $2y$

"x divided by 4": $\frac{x}{4}$

In $3x + 5$: the coefficient is 3, the constant is 5

2 Substituting

Replace each letter with its number, then work it out.

$x = 4$: $3x + 2 = 14$ $x^2 = 16$

$20 \div x = 5$

$a = 2,\ b = 5$: $ab + 1 = 11$

Watch out: $3x$ with $x = 4$ is 3×4, not 34.

3 Equivalent expressions

Equivalent expressions are equal for every value of the letter.

$2(x + 3)$ and $2x + 6$

Test with $x = 5$: both give 16 ✓

$x + x = 2x$

$3a + 2a = 5a$

4 Like terms

Only terms with the same letters and powers can be added or subtracted.

$3x + 2x = 5x$

$4a + 2b - a = 3a + 2b$

$5x$ and x^2 aren't like terms

$2x + 3 + x + 4 = 3x + 7$

5 Multiplying and dividing

Multiply the numbers, then the letters. Divide like a fraction.

$3 \times 2x = 6x$ $2a \times 3b = 6ab$

$x \times x = x^2$

$8x \div 2 = 4x$ $12ab \div 4a = 3b$

6 Expanding brackets

Multiply everything inside by the term outside.

$3(x + 2) = 3x + 6$

$5(2a - 1) = 10a - 5$

$x(x + 4) = x^2 + 4x$

Watch out: multiply the second term too.

7 Applying algebra

Pick a letter for the unknown and write a rule.

Gym: \$10 to join plus \$5 a visit: $C = 10 + 5v$. 8 visits: \$50

Rectangle, width x, length $x + 2$: perimeter $4x + 4$. If $x = 3$, that's 16.

30 students on each of b buses: $30b$. 4 buses: 120 students